PRODUCT DATASHEET

DIGITAL MOTOR CHECKER

MCM Instruments | Model EMC-38

Nayara RFQ: 6000123754 | **Item:** 00010 | **Mat Code:** 6784140114 | **Qty:** 2 Nos

OVERVIEW

Portable digital motor checker for testing rotor and stator windings of electric motors – measures winding resistance, inductance and insulation resistance in one instrument, for quick fault diagnosis (shorted turns, open windings, insulation breakdown) during maintenance.

KEY FEATURES

- Rotor & stator winding resistance measurement down to 0.1 mΩ resolution
- Inductance measurement up to 2000 mH for detection of inter-turn faults
- Insulation resistance test at 500 V DC up to 200 MΩ
- Rechargeable battery pack with charger for field use
- Rugged portable design with carry case

TECHNICAL SPECIFICATIONS

Parameter	Specification
Insulation resistance	0 – 200 MΩ at 500 V DC, ±(2% + 1 digit), resolution 0.1 MΩ
Resistance – 200 mΩ range	Resolution 0.10 mΩ, accuracy ±2%
Resistance – 2 Ω range	Resolution 0.001 Ω, accuracy ±2%
Resistance – 20 Ω range	Resolution 0.01 Ω, accuracy ±2%
Resistance – 200 Ω range	Resolution 0.1 Ω, accuracy ±2%
Inductance – 2 mH range	Resolution 0.001 mH, accuracy ±3%
Inductance – 20 mH range	Resolution 0.01 mH, accuracy ±3%
Inductance – 200 mH range	Resolution 0.1 mH, accuracy ±3%
Inductance – 2000 mH range	Resolution 1 mH, accuracy ±3%
Display	Digital
Power supply	Rechargeable battery pack with charger

SCOPE OF SUPPLY (PER UNIT)

• EMC-38 Digital Motor Checker – 1 no. • Rechargeable battery pack with charger • Test leads • Carry case • Instruction manual • Test & calibration report, warranty / guarantee certificate

COMPLIANCE TO RFQ SPECIFICATION

RFQ requirement	Offered	Status
Type: Digital motor checker	MCM EMC-38 digital motor checker	**Complied**
Make / Model: MCM Instruments EMC-38	MCM Instruments EMC-38	**Complied**
IR: 0–200 MΩ at 500 V DC ±(2%+1 digit), 0.1 MΩ res.	As specified	**Complied**
Resistance 200 mΩ / 2 Ω / 20 Ω / 200 Ω, ±2%	As specified	**Complied**
Inductance 2 / 20 / 200 / 2000 mH, ±3%	As specified	**Complied**
Rechargeable battery pack with charger	Supplied	**Complied**
Test & calibration report, warranty certificate, manual, carry case	Supplied	**Complied**

Warranty: 18 months from installation or 24 months from delivery, whichever is earlier (Syselec contract warranty). Specifications are as per OEM published data and the RFQ; refer to the enclosed OEM catalogue. Specifications subject to change by OEM without notice.

Regional Head Office: Syselec Technologies Pvt. Ltd., A wing, 305-306, Vishnu Shivam Tower, Behind Vishnu Shivam Mall, Thakur Village, Kandivli East, Mumbai-400101. Tel: +918108123468/61 | md@syselec.net

System Integration Work Address:
Syselec Works, Veer Paras Estate, D-7/5, Indira Nagar, Wagle Industrial Estate, Thane West, Maharashtra 400 604

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PRODUCT DATASHEET

DIGITAL VIBRATION PEN

Kusam-Meco | Model KM-63

Nayara RFQ: 6000123754 | **Item:** 00020 | **Mat Code:** 9823341494 | **Qty:** 4 Nos

OVERVIEW

Pocket-size pen-type vibration meter for condition monitoring of rotating machinery – pumps, motors, fans and compressors. Measures vibration acceleration, velocity and displacement with a high-sensitivity piezoelectric probe for quick route-based checks.

KEY FEATURES

- Measures acceleration, velocity and displacement
- Selectable measurement frequency bands (LO 10 Hz–1 kHz / HI 1–15 kHz)
- High-sensitivity shear-type piezoelectric ceramic accelerometer
- Long and short probe heads for different measuring locations
- AC signal output for earphone / analyser connection
- 3½-digit backlit LCD, low-battery indication and auto power-off
- Compact, portable, simple to use

TECHNICAL SPECIFICATIONS

Parameter	Specification
Vibration pick-up	Piezoelectric ceramic accelerometer (shear type)
Acceleration range	0.1 – 199.9 m/s² peak
Velocity range	0.1 – 199.9 mm/s RMS
Displacement range	0.001 – 1.999 mm p-p
Accuracy	±(5% + 2 digits)
Frequency range – acceleration	10 Hz – 1 kHz (LO), 1 kHz – 15 kHz (HI)
Frequency range – velocity / displacement	10 Hz – 1 kHz (LO)
Display / update	3½-digit backlit LCD / 1 second
Signal output	AC output 2 V peak (full scale), load impedance ≥ 10 kΩ
Current consumption	Static ≤ 20 mA, operating ≤ 25 mA
Power supply / battery life	9 V battery / approx. 20 h continuous
Low battery indication / auto power-off	6.9 V ± 0.2 V / after 60 s
Operating conditions	0 – 40 °C, 30 – 90% RH
Size / weight	67 × 30 × 183 mm / 182 g (incl. battery)

SCOPE OF SUPPLY (PER UNIT)

• KM-63 Vibration Pen – 1 no. • Long and short probe heads • 9 V battery • Instruction manual • Test certificate, warranty certificate

COMPLIANCE TO RFQ SPECIFICATION

RFQ requirement	Offered	Status
Type: Vibration pen, Make/Model Kusam-Meco KM-63	Kusam-Meco KM-63	**Complied**
Acceleration 0.1–199.9 m/s², velocity 0.1–199.9 mm/s, displacement 0.001–1.999 mm	As specified	**Complied**
Accuracy ±(5% + 2 digits)	As specified	**Complied**
Frequency 10 Hz–1 kHz (LO), 1–15 kHz (HI)	As specified	**Complied**
3.5-digit backlit LCD, AC output, auto power-off, low-battery indication	As specified	**Complied**
9 V battery, ~20 h life, 182 g, 67 × 30 × 183 mm	As specified	**Complied**

Warranty: 18 months from installation or 24 months from delivery, whichever is earlier (Syselec contract warranty). Specifications are as per OEM published data and the RFQ; refer to the enclosed OEM catalogue. Specifications subject to change by OEM without notice.

PRODUCT DATASHEET

DIGITAL TRUE-RMS MULTIMETER

Fluke | Model 179

Nayara RFQ: 6000123754 | **Item:** 00030 | **Mat Code:** 6785140363 | **Qty:** 4 Nos

OVERVIEW

Industrial-grade true-RMS digital multimeter for accurate measurement on non-linear loads in plant electrical maintenance. 6000-count display with analog bargraph, built-in temperature measurement and safety-rated for CAT III 1000 V / CAT IV 600 V environments.

KEY FEATURES

- True-RMS AC voltage and current
- Basic DC accuracy 0.09%
- Analog bargraph, large backlit display
- Auto / manual ranging, Min / Max / Average, Display Hold
- Continuity beeper and diode test
- Built-in thermometer with K-type probe
- Safety: CAT III 1000 V, CAT IV 600 V

TECHNICAL SPECIFICATIONS

Parameter	Specification
DC / AC voltage	Up to 1000 V; DC accuracy ±(0.09% + 2)
DC / AC current	Up to 10 A (20 A for 30 s)
Resistance	Up to 50 MΩ
Capacitance	Up to 10,000 µF
Frequency	Up to 100 kHz
Temperature	−40 °C to +400 °C (with 80BK-A probe)
Display	6000 counts, analog bargraph, backlight
Functions	Continuity, diode test, Min/Max/Avg, Hold, auto/manual range
Operating temperature	−10 °C to +50 °C
Safety rating	IEC/EN 61010-1, CAT III 1000 V / CAT IV 600 V
Approvals	CE, CSA, UL, TÜV
Power	9 V alkaline battery
OEM warranty	Fluke limited lifetime warranty

SCOPE OF SUPPLY (PER UNIT)

• Fluke 179 True-RMS Multimeter – 1 no. • TL75 test leads (test probes) • 80BK-A integrated DMM temperature probe • Removable holster • 9 V battery (installed) • User manual • OEM test / calibration certificate

COMPLIANCE TO RFQ SPECIFICATION

RFQ requirement	Offered	Status
True-RMS, voltage AC/DC 1000 V	True-RMS, up to 1000 V AC/DC	**Complied**
Basic DC accuracy 0.09%	±(0.09% + 2)	**Complied**
Current AC/DC 10 A; resistance 50 MΩ	As specified	**Complied**
Frequency 100 kHz; capacitance 10,000 µF	As specified	**Complied**
Continuity / diode, analog bargraph, backlight, auto/manual, Min/Max/Hold	Available	**Complied**
Non-contact voltage detection	Not available on Fluke 179 (model named in RFQ)	**Deviation**
Operating temperature −10 °C to +50 °C	As specified	**Complied**
EN 61010-1 CAT III 1000 V / CAT IV 600 V; CE	As specified	**Complied**
Accessories: removable holster, test probes	Supplied	**Complied**
Warranty 3 years	Fluke limited lifetime + Syselec 18/24-month contract warranty	**Complied**
Calibration / test certificate	Supplied	**Complied**

Warranty: 18 months from installation or 24 months from delivery, whichever is earlier (Syselec contract warranty). Specifications are as per OEM published data and the RFQ; refer to the enclosed OEM catalogue. Specifications subject to change by OEM without notice.

PRODUCT DATASHEET

DIGITAL INSULATION TESTER

Fluke | Model 1507

Nayara RFQ: 6000123754 | **Item:** 00040 | **Mat Code:** 6784140074 | **Qty:** 2 Nos

OVERVIEW

Compact, rugged insulation resistance tester for troubleshooting and preventive maintenance of motors, cables, switchgear and transformers. Multiple test voltages, automatic PI / DAR, earth-bond resistance and live-circuit detection, with remote probe for one-handed testing.

KEY FEATURES

- Insulation test voltages 50, 100, 250, 500 and 1000 V
- Insulation resistance 0.01 MΩ to 10 GΩ
- Automatic Polarisation Index (PI) and Dielectric Absorption Ratio (DAR)
- Earth-bond / low-resistance measurement (200 mA)
- Live-circuit detection inhibits test if voltage > 30 V present
- Automatic discharge of capacitive voltage; pass/fail compare function
- Remote probe for one-handed testing; CAT IV 600 V

TECHNICAL SPECIFICATIONS

Parameter	Specification
Insulation resistance range	0.01 MΩ to 10 GΩ
Insulation test voltages	50 / 100 / 250 / 500 / 1000 V
AC/DC voltage measurement	0.1 V to 600 V
Max. voltage on any terminal	600 V AC RMS or DC
Operating / storage temperature	−20 °C to 55 °C / −40 °C to 60 °C
Relative humidity (non-condensing)	0–95% @ 10–30 °C; 0–75% @ 30–40 °C; 0–40% @ 40–55 °C
Vibration / shock	Random 2 g, 5–500 Hz (MIL-PRF-28800F Class 2) / 1 m drop
EMC	EN 61326-1, accuracy as specified in 3 V/m RF field
Safety	IEC/EN 61010-1, CAT IV 600 V; CSA, TÜV certified
Batteries / life	4 × AA; ≥ 1000 insulation tests, ≥ 2500 earth-bond tests
Size / weight	5.0 × 10.0 × 20.3 cm / 550 g
IP rating / altitude	IP40 / 2000 m CAT IV 600 V, 3000 m CAT III 600 V
Over-range capability	110% of range

SCOPE OF SUPPLY (PER UNIT)

• Fluke 1507 Insulation Tester – 1 no. • TL224 test leads, TP74 test probes • Alligator clips – red (PN 1958654) and black (PN 1958646) • Remote probe and holster • C101 hard carrying case • 4 × AA batteries, user manual • OEM test certificate

COMPLIANCE TO RFQ SPECIFICATION

RFQ requirement	Offered	Status
Range 0.01 MΩ to 10 GΩ	As specified	**Complied**
Voltage 0.1 V to 600 V; max 600 V AC RMS/DC on any terminal	As specified	**Complied**
Environmental: −20 to 55 °C operating; humidity, vibration, shock	As specified	**Complied**
EMC EN 61326-1; Safety CAT IV 600 V; CSA / TÜV	As specified	**Complied**
4 × AA batteries; ≥ 1000 IR / ≥ 2500 earth-bond tests	As specified	**Complied**
Size 5.0 × 10.0 × 20.3 cm, 550 g, IP40	As specified	**Complied**
Accessories: TL224, TP74, clips, holster, remote probe, C101 hard case	Supplied	**Complied**
Test certificate	Supplied	**Complied**

Warranty: 18 months from installation or 24 months from delivery, whichever is earlier (Syselec contract warranty). Specifications are as per OEM published data and the RFQ; refer to the enclosed OEM catalogue. Specifications subject to change by OEM without notice.

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